

AI · ML · DATA SCIENCE

Repository Guidelines

Source: AGENTS.md

■ Project Overview

This repository is a comprehensive collection of tutorials, practice exercises, and utilities covering AI, Machine Learning, Data Science, and modern software engineering tools. It includes hands-on materials for Python, PySpark, SQL, Go, Docker, LangChain, and LlamaIndex.

■ Project Structure & Module Organization

The repository is organized into numbered directories, each dedicated to a specific topic:

- **Python & DS Basics:** `1.Python` , `3.SQL` , `4.Theory`
- **Big Data & Infrastructure:** `2.PySpark` , `11.Docker` , `14.Databricks`
- **LLM & AI Orchestration:** `20.LlamaIndex` , `23.LangChain` , `24.LangChain2`
- **Utilities & Tooling:** `18.Utilities` contains various helper scripts like `images_to_pdf.py` and `send-mail.py`.
- **Tutorials:** `25.Claude-Code-Deep-Dive` contains specific agent-related learning materials.

■ Build, Test, and Development Commands

The repository uses multiple dependency management systems depending on the module.

Main Python Environment

- **Install dependencies:** `pipenv install` or `pip install -r requirements.txt`
- **Python version:** 3.11 (as specified in `Pipfile`)

Sub-module Environments

- **uv (15.uv_demo):** Uses `uv` for fast environment management.
- **Pipenv:** Several directories (e.g., `20.LlamaIndex` , `23.LangChain`) have their own `Pipfile` .
- **Go (19.Go):** Standard Go commands like `go run <file>.go` .

■ Coding Style & Naming Conventions

- **Language Dominance:** Python is the primary language, often used in Jupyter Notebooks (`.ipynb`) for interactive learning.
- **VS Code Integration:** Recommended settings are stored in `.vscode/settings.json` , with `python.terminal.activateEnvironment` disabled to allow manual control.

■ Commit & Pull Request Guidelines

- **Commit Messages:** Use concise, descriptive messages. Common patterns in the repo include “added content” or “Add [topic] setup”.
- **Workflow:** Direct commits are common for adding new learning materials. No specific PR template is enforced.